

🚩 Flag 1 – Install Flutter SDK

Objective: Install the Flutter SDK.

Verify:

flutter --version

Screenshot:

- The results of the command

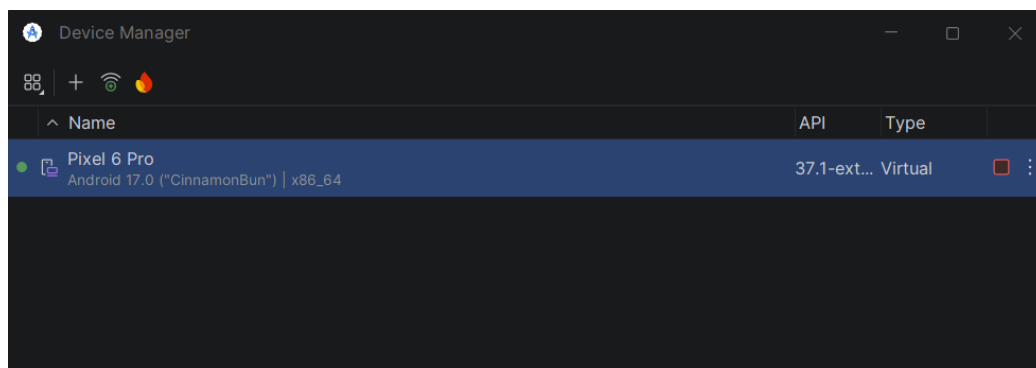
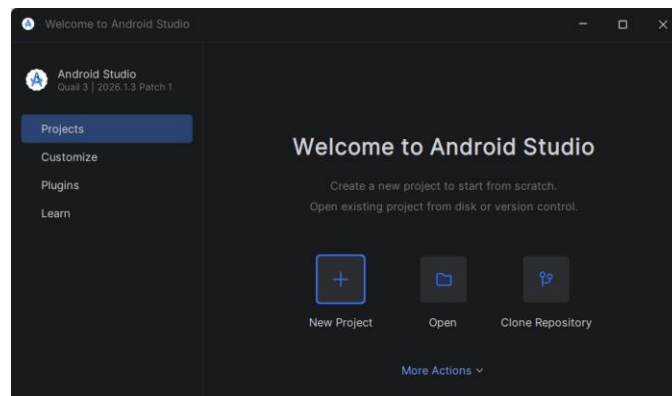
```
C:\Users\admin>flutter --version
Flutter 3.44.9 • channel stable • https://github.com/flutter/flutter.git
Framework • revision 6b182d2c75 (6 days ago) • 2026-08-05 10:04:07 -0700
Engine • hash b9499e4c25212536ba3a4eec4f5c1905fb3214fe (revision 5a2a6a42cc) (10 days ago) • 2026-07-31 18:31:59.000Z
Tools • Dart 3.12.2 • DevTools 2.57.0
```

🚩 Flag 2 – Install Android Studio

Objective: Install Android Studio.

Screenshot:

- Android Studio installed
- SDK Manager opened (or Welcome screen)



🚩 Flag 3 – Install VS Code & Flutter Extension

Objective: Install VS Code and the Flutter (which also installs Dart) extension.

Screenshot:

- Flutter extension installed
- Dart extension installed (or shown as installed)



🚩 Flag 4 – Run Flutter Doctor

Objective: Check whether everything is configured correctly.

Run:

flutter doctor

Screenshot:

- flutter doctor results

```
C:\Users\admin>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.44.9, on Microsoft Windows [Version 10.0.26200.8973], locale en-US)
[✓] Windows Version (Windows 11 or higher, 25H2, 2009)
[!] Android toolchain - develop for Android devices (Android SDK version 36.0.0)
    ! Some Android licenses not accepted. To resolve this, run: flutter doctor --android-licenses
[✓] Chrome - develop for the web
[!] Visual Studio - develop Windows apps (Visual Studio Community 2022 17.14.29 (March 2026))
    X Visual Studio is missing necessary components. Please re-run the Visual Studio installer for the "Desktop
      development with C++" workload, and include these components:
        MSVC v142 - VS 2019 C++ x64/x86 build tools
          - If there are multiple build tool versions available, install the latest
        C++ CMake tools for Windows
        Windows 10 SDK
[✓] Connected device (4 available)
[✓] Network resources

! Doctor found issues in 2 categories.

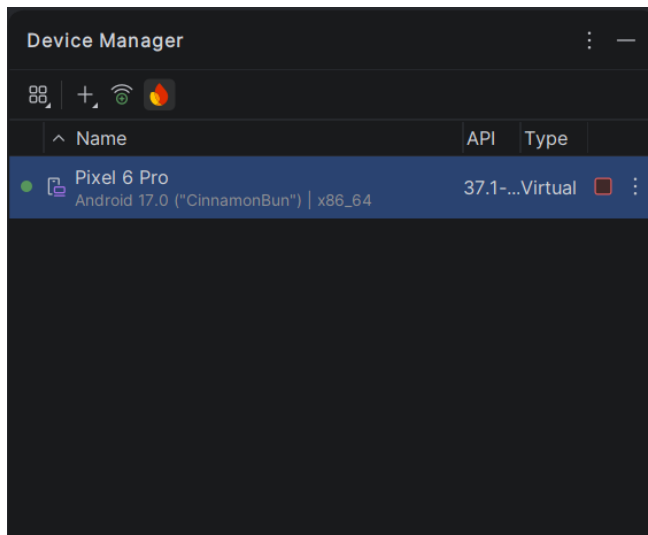
C:\Users\admin>
```

🚩 Flag 5 – Build Your Virtual Device

Objective: Create your first Android Emulator (AVD).

Screenshot:

- Device Manager
- Your newly created virtual device
- Emulator successfully booted



🚩 Flag 6 – Forge Your Project

Objective: Create your first Flutter project.

Run:

```
flutter create my_first_flutter_app
```

Screenshot:

- Successful creation of project

```
C:\Users\admin\Documents\FlutterProjects>flutter create my_first_flutter_app
Creating project my_first_flutter_app...
Resolving dependencies in 'my_first_flutter_app'... (1.2s)
Downloading packages...
Got dependencies in 'my_first_flutter_app'.
Wrote 131 files.

All done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev

In order to run your application, type:

$ cd my_first_flutter_app
$ flutter run

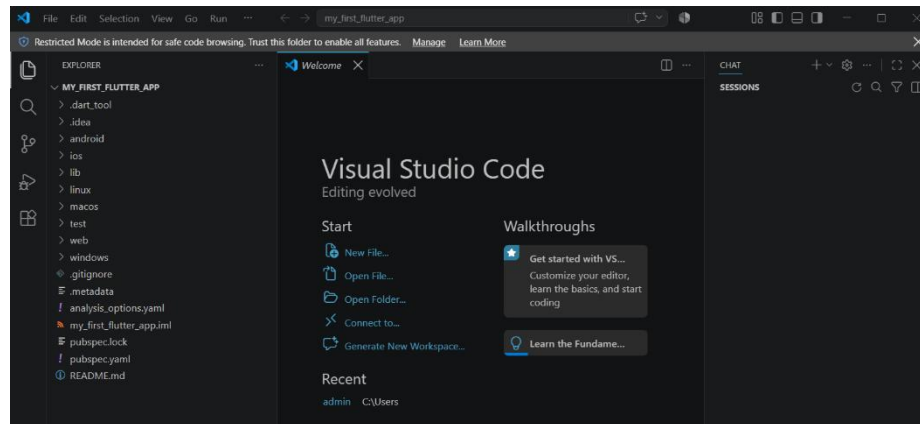
Your application code is in my_first_flutter_app\lib\main.dart.
```

🚩 Flag 7 – Unlock the Workspace

Objective: Open the project in VS Code.

Screenshot:

- Your project opened in VS Code



🚩 Flag 8 – Launch

Objective: Run your Flutter application.

Run:

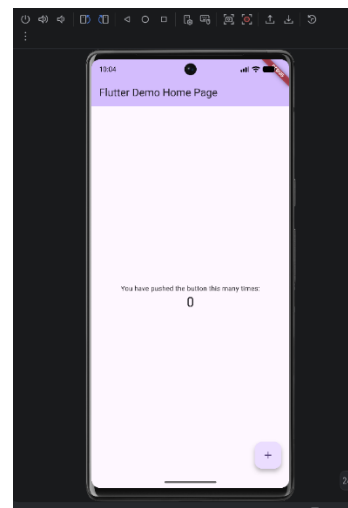
flutter run

Screenshot:

- Terminal
- Emulator running
- Default Flutter app

```
PS C:\Users\admin\Documents\FlutterProjects\my_first_flutter_app> flutter run
Launching lib\main.dart on sdk gphone16k x86_64 in debug mode...
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by net.rubygrapefruit.platform.internal.NativeLibraryLoader in an unna
med module (file:/C:/Users/admin/.gradle/wrapper/dists/gradle-9.1.0-all/7wzd0jkjite6iaq2p43wpjgi9/gradle-9.1.0/lib/nat
ive-platform-0.22-milestone-28.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

Checking the license for package NDK (Side by side) 28.2.13676358 in C:\Users\admin\AppData\Local\Android\Sdk\licenses
License for package NDK (Side by side) 28.2.13676358 accepted.
Preparing "Install NDK (Side by side) 28.2.13676358 v.28.2.13676358".
"Install NDK (Side by side) 28.2.13676358 v.28.2.13676358" ready.
Installing NDK (Side by side) 28.2.13676358 in C:\Users\admin\AppData\Local\Android\Sdk\ndk\28.2.13676358
"Install NDK (Side by side) 28.2.13676358 v.28.2.13676358" complete.
"Install NDK (Side by side) 28.2.13676358 v.28.2.13676358" finished.
Checking the license for package Android SDK Platform 36 in C:\Users\admin\AppData\Local\Android\Sdk\licenses
License for package Android SDK Platform 36 accepted.
Preparing "Install Android SDK Platform 36 (revision 2)".
"Install Android SDK Platform 36 (revision 2)" ready.
Installing Android SDK Platform 36 in C:\Users\admin\AppData\Local\Android\Sdk\platforms\android-36
"Install Android SDK Platform 36 (revision 2)" complete.
"Install Android SDK Platform 36 (revision 2)" finished.
[Progress bar] 25%
Checking the license for package CMake 3.22.1 in C:\Users\admin\AppData\Local\Android\Sdk\licenses
License for package CMake 3.22.1 accepted.
Preparing "Install CMake 3.22.1 v.3.22.1".
"Install CMake 3.22.1 v.3.22.1" ready.
Installing CMake 3.22.1 in C:\Users\admin\AppData\Local\Android\Sdk\cmake\3.22.1
"Install CMake 3.22.1 v.3.22.1" complete.
```



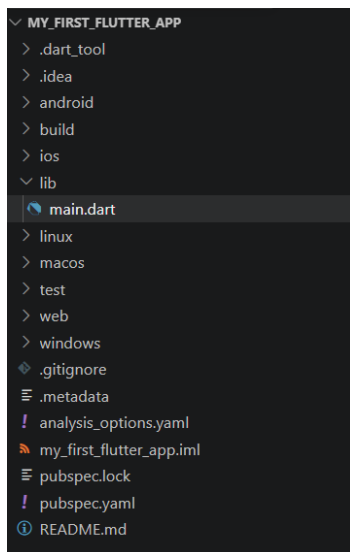
^ Name	API	Type
  Device at 192.168.142.181:37999 Check for device software updates to improve Wi-Fi pairing.		Physical  
 Pixel 6 Pro Android 17.0 ("CinnamonBun") x86_64	37.1-e... Virtual	 

🚩 Flag 9 – Find the Entry Point

Figure out where the default Flutter page comes from.

Screenshot:

- The location of the file where the default page is from (the one showing on the emulator)



```
lib > main.dart > ...
1  import 'package:flutter/material.dart';
2
3  Run | Debug | Profile
4  void main() {
5    runApp(const MyApp());
6  }
7
8  class MyApp extends StatelessWidget {
9    const MyApp({super.key});
10
11    // This widget is the root of your application.
12    @override
13    Widget build(BuildContext context) {
14      return MaterialApp(
15        title: 'Flutter Demo',
16        theme: ThemeData(
17          // This is the theme of your application.
```

```
38  class MyHomePage extends StatefulWidget {
39    const MyHomePage({super.key, required this.title});
40
41    // This widget is the home page of your application. It is stateful, meaning
42    // that it has a State object (defined below) that contains fields that affect
43    // how it looks.
44
45    // This class is the configuration for the state. It holds the values (in this
46    // case the title) provided by the parent (in this case the App widget) and
47    // used by the build method of the State. Fields in a Widget subclass are
48    // always marked "final".
49
50    final String title;
51
52    @override
53    State<MyHomePage> createState() => _MyHomePageState();
54  }
55
56  class _MyHomePageState extends State<MyHomePage> {
57    int _counter = 0;
58
59    void _incrementCounter() {
60      setState(() {
```

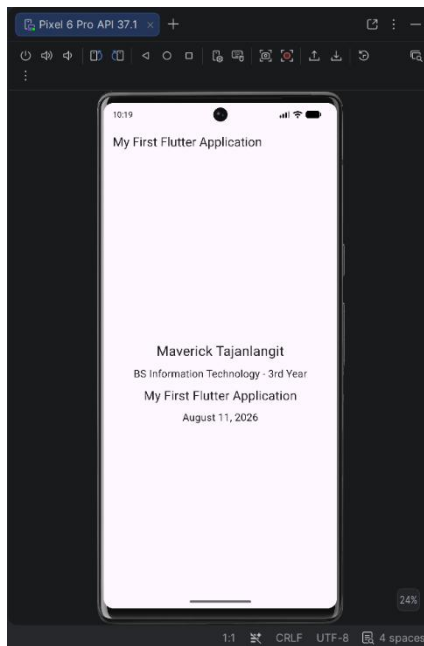
🚩 Flag 10 – Claim Your App

Replace the default screen with:

- Full Name
- Course & Section
- "My First Flutter Application"
- Today's Date

Screenshot:

- Your updated content of the code
- The updated UI



```
lib > main.dart > MyApp > build
1 import 'package:flutter/material.dart';
2
3 Run [Debug] Profile
4 void main() {
5   runApp(const MyApp());
6 }
7
8 class MyApp extends StatelessWidget {
9   const MyApp({super.key});
10
11 @override
12 Widget build(BuildContext context) {
13   return MaterialApp(
14     debugShowCheckedModeBanner: false,
15     home: Scaffold(
16       appBar: AppBar(
17         title: const Text('My First Flutter Application'),
18       ), // AppBar
19       body: Center(
20         child: Column(
21           mainAxisAlignment: MainAxisAlignment.center,
22           children: const [
23             Text(
24               'Maverick Tajanlangit',
25               style: TextStyle(fontSize: 24),
26             ),
27           ],
28         ),
29       ), // Center
30     ), // Scaffold
31   );
```

```
27   SizedBox(height: 10),
28
29   Text(
30     'BS Information Technology - 3rd Year',
31     style: TextStyle(fontSize: 18),
32   ), // Text
33
34   SizedBox(height: 10),
35
36   Text(
37     'My First Flutter Application',
38     style: TextStyle(fontSize: 22),
39   ), // Text
40
41   SizedBox(height: 10),
42
43   Text(
44     'August 11, 2026',
45     style: TextStyle(fontSize: 18),
46   ), // Text
47 ],
48 ), // Column
49 ), // Center
50 ), // Scaffold
```

🚩 Flag 11 – Personalize It

Research and implement at least **three** visual changes.

Examples:

- Change the background colour.
- Change the text colour.
- Centre the content.
- Increase font size.
- Add a border or rounded card.
- Change the font.
- Add an emoji or icon.

Screenshot:

- The code changed for personalizing
- The updated UI



```
10 @override
11 widget build(BuildContext context) {
12   return MaterialApp(
13     debugShowCheckedModeBanner: false,
14     home: Scaffold(
15       backgroundColor: const Color(0xFF50C878), // Emerald green
16
17       body: Center(
18         child: Column(
19           mainAxisAlignment: MainAxisAlignment.center,
20           children: [
21             // Leaf icon
22             const Text(
23               '🌿',
24               style: TextStyle(fontSize: 45),
25             ), // Text
26
27             const SizedBox(height: 20),
28
29             // Full Name
30             const Text(
31               'Maverick Tajanlangit',
32               maxLines: 1,
33               style: TextStyle(
34                 color: colors.white,
```

```
35               fontSize: 30,
36               fontWeight: FontWeight.bold,
37               fontStyle: FontStyle.italic,
38             ), // TextStyle
39           ], // Text
40
41           const SizedBox(height: 15),
42
43           // Course & Section
44           const Text(
45             'BS Information Technology - 2nd Year',
46             textAlign: TextAlign.center,
47             style: TextStyle(
48               color: colors.white,
49               fontSize: 19,
50             ), // TextStyle
51           ), // Text
52
53           const SizedBox(height: 15),
54
55           // Application Title
56           const Text(
57             'My First Flutter Application',
58             textAlign: TextAlign.center,
```

```

59         style: TextStyle(
60           color: Colors.white,
61           fontSize: 22,
62           fontWeight: FontWeight.w600,
63         ), // TextStyle
64       ), // Text
65     ),
66     const SizedBox(height: 15),
67     // Date
68     const Text(
69       'August 11, 2026',
70       style: TextStyle(
71         color: Colors.white,
72         fontSize: 18,
73       ), // TextStyle
74     ), // Text
75   ],
76 ), // Column
77 ), // Center
78 ), // Scaffold
79 ); // MaterialApp
80 }
81 }
82 }

```

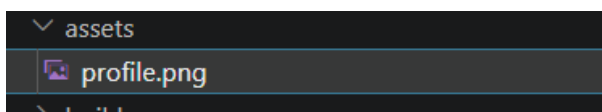
🚩 Flag 12 – Add an Image

Research how to:

- Create an assets folder
- Register assets
- Display an image

Screenshot:

- The steps it took to add an image
- The updated UI



```

! pubspec.yaml
38 dev_dependencies:
39   flutter_test:
40     sdk: flutter
41
42 flutter_lints: ^6.0.0
43
44 # For information on the generic Dart part of this file, see the
45 # following page: https://dart.dev/tools/pub/pubspec
46
47 # The following section is specific to Flutter packages.
48 flutter:
49
50   # The following line ensures that the Material Icons font is
51   # included with your application, so that you can use the icons in
52   # the material Icons class.
53   uses-material-design: true
54
55   assets:
56     - assets/profile.png
57
58
59
60
61
62

```



```
// Profile Image
clipOval(
  child: Image.asset(
    'assets/profile.png',
    width: 120,
    height: 120,
    fit: BoxFit.cover,
  ), // Image.asset
), // ClipOval

const SizedBox(height: 25),
```

